SERVER



Advancing Arm Server System Architecture in the OCP Ecosystem

SYSTEMS MANAGEMENT



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OCP PLATINUM® Member

Arm SystemReady band

Defined in Arm SystemArchAC

Contribute to OCP and further development in OCP Server and Hardware Management Projects

Arm owned open source

Server: Common server management interfaces such as Redfish, IPMI, MCTP, PLDM, SPDM

Server Base Manageability Requirements (SBMR)

Windows

Install, boot and run on standard OS and hypervisors

Base Boot Security Requirements (BBSR)

SBBR

Base Boot Requirements (BBR)

Common firmware interfaces such as UEFI/ACPI/SMBIOS/, including secure boot and firmware updates. TPM required for physical servers

Server Base System Architecture (SBSA)

Server: Processor & Server System Architecture physical hardware standardization

Base System Architecture (BSA) (Physical or Virtual)

Baseline hardware requirements for Arm platform architecture

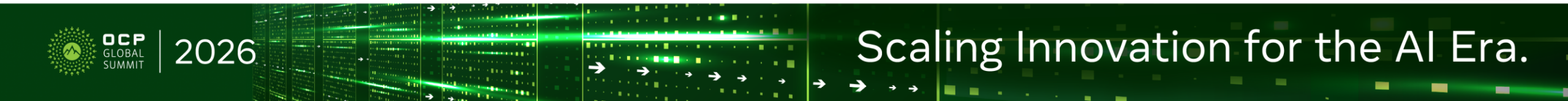
Architecture Compliance Suite (**ACS**) Testing

arm SystemReady

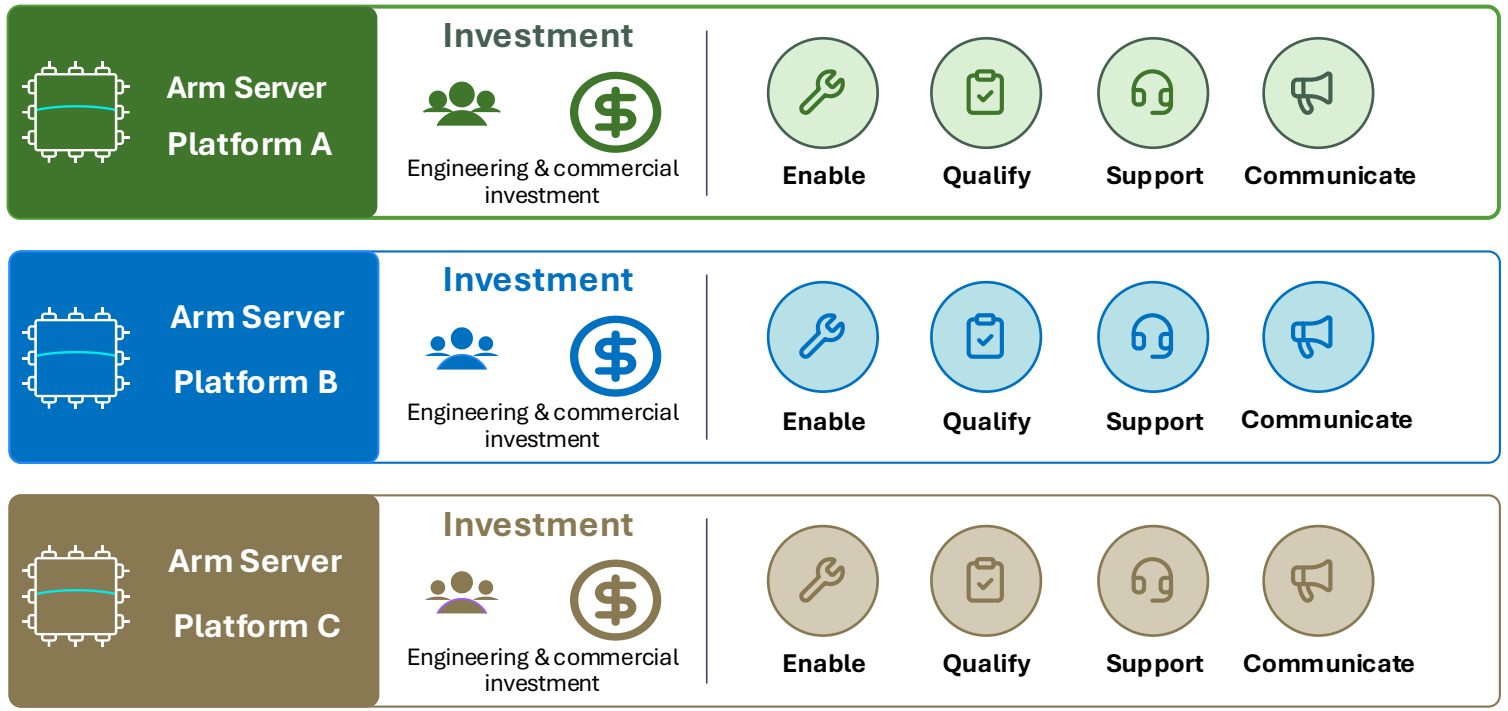
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Platform-by-Platform Enablement Caps the Opportunity



Three platforms.

Three separate enablement investments.

Each platform requires dedicated engineering and commercial investment cycle.

Limited reuse

Enablement assets, evidence, and knowledge remain platform-specific.

Limited leverage

Each investment reaches a narrow platform and customer set.

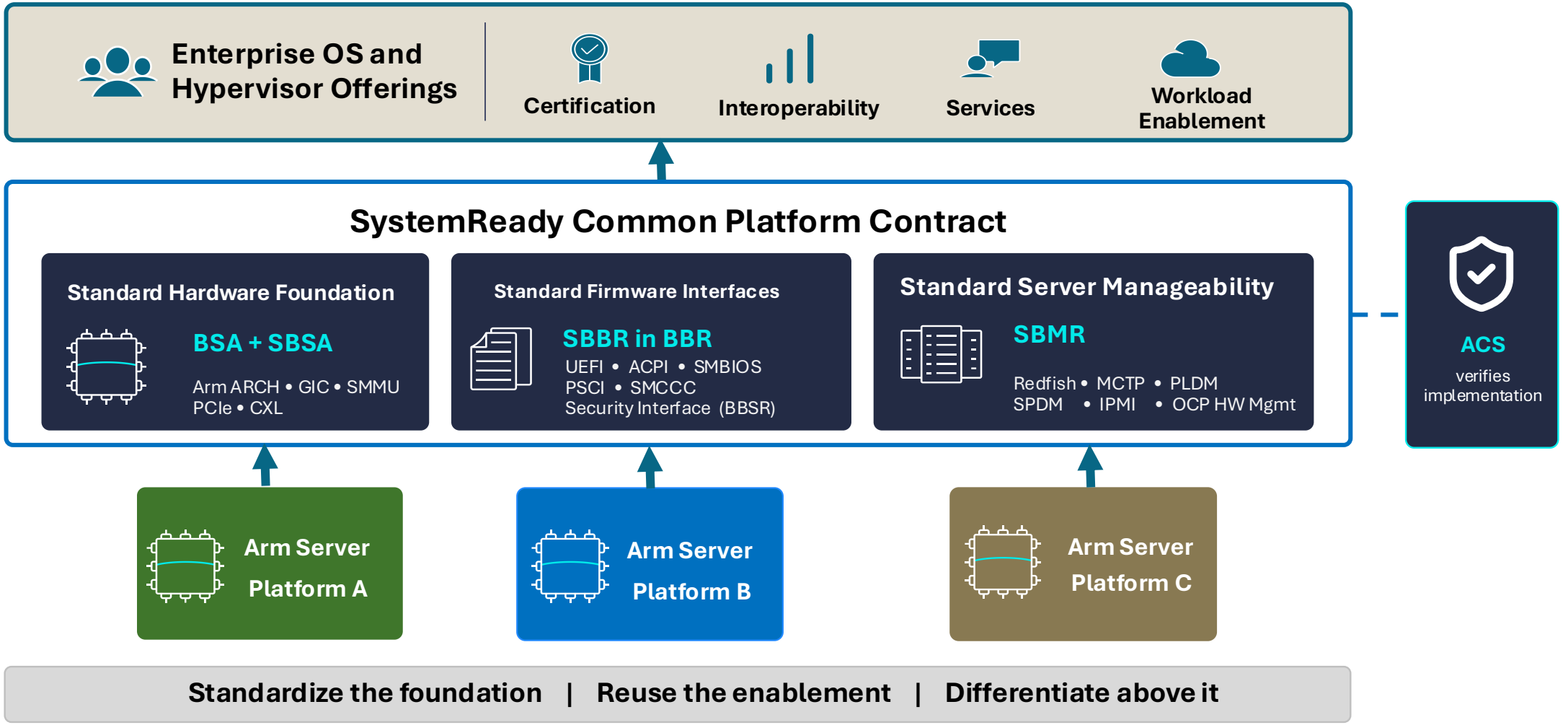
Slower expansion

The next platform must rebuild readiness and market confidence.

More platforms should multiply market reach - not multiply duplicated investment.

How do we turn **one-off enablement** into **reusable ecosystem value**?

SystemReady Creates a Common Platform Contract



Clean Starting Point for OSVs

1 Shift-Left for Readiness

SiPs • ODMs • OEMs



- Pre-silicon validation
- Platform & firmware bring-up
- Release regression

Fewer platform issues reach OSV qualification

Platform partners run ACS. OSVs consume the evidence

2 Low-Friction Execution

ACS

Architecture Compliance Suites

- Prebuilt live image or standalone modules
- Automated UEFI + Linux workflow
- Runs across standard UEFI firmware and Linux environments
- Standardized JSON + HTML reports

3 A Clear Readiness Gate



Baseline Ready

Proceed to OS qualification

Action Required

Return findings to the platform owner

Summary first. Detailed evidence when needed.

4 OSVs Differentiate and Scale



- Focus on OS features and performance
- Qualify more platforms and serve more customers
- Expand support, certification, and service opportunities

Less Platform Debugging. More Capacity to Grow

Platform partners validate the baseline. OSVs differentiate and scale across more platforms and customers. ACS complements OS certification - it does not replace it.

Replicate a Proven Platform-to-Market Model

PROVEN MODEL

NVIDIA

Grace Server



A clean starting point for OSVs

1. SystemReady Required across Platforms
2. OEM/ODM Achieves SystemReady Compliance

Reusable OSV enablement

3. OSV Qualification + Support

Market Adoption

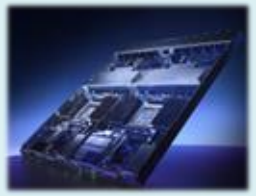
4. Enterprise Server Deployment

A Proven Path to Market

APPLY THE MODEL

arm

AGI CPU SERVER



A clean starting point for OSVs

1. Arm AGI CPU Reference Server is SystemReady SR Band compliant (at SBSA Level 6!)
2. OEM/ODM Achieves SystemReady Compliance

Reusable OSV enablement

3. OSV Qualification + Support

Market Adoption

4. Enterprise Server Adoption

Faster Readiness. Broader Opportunity

Reuse the proven model – Not another platform-specific enablement path

OCP Turns a Platform Baseline into an Industry Baseline

Connect technical baselines to real-world adoption.

Arm-Developed Platform Baseline

SBSA + SBMR

Arm-based server architecture and manageability requirements

Arm contributes the baseline

Neutral Governance

Structured multi-company review and decision-making.

- Common process
- Shared roadmap
- Foundation oversight

Industry Ownership

The ecosystem helps shape priorities and requirements.

CSPs · OEMs / SiPs · OSVs · FW Vendors

Adoption Connection

Connect the baseline to how systems are selected and deployed.

Design → Qualification → Deployment → Procurement

Shared Industry Baseline

- Broader ecosystem influence
- Stronger alignment
- Closer connection to deployment needs
- A more durable adoption path

Contributing SBSA and SBMR to OCP

- Arm SystemReady band is required for OCP-Accepted Arm-based servers
  - See Hardware Specification at https://www.opencompute.org/contributions/templates-agreements



- Arm contributed the SBSA and SBMR specifications to OCP at the 2026 OCP EMEA summit
  - SBSA public wiki: https://github.com/opencomputeproject/ocp-sbmr/wiki
  - SBSA public mailing list: https://ocp-all.groups.io/g/ocp-sbmr-cla
  - SBMR public wiki: https://github.com/opencomputeproject/ocp-sbmr/wiki
  - SBMR public mailing list: https://ocp-all.groups.io/g/ocp-sbmr-cla

OCP SBSA CLA Workstream Members

ami

AMPERE.

arm

Google Cloud

GRAPHCORE

insyde

Meta

Microsoft

NVIDIA

SUSE

Canonical

Red Hat

Joint OCP recently!

Extending invitation to join SBSA CLA Workstream

SERVER



SBSA Specification Update

- **Arm specifications**
  - SBSA 3.1 (Feb 2016)
  - SBSA 7.0 (Jan 2021)
  - Current public release: SBSA 8.0 (Jul 2025) https://developer.arm.com/documentation/den0029/latest/
- **OCP Specifications**
  - OCP SBSA 8.1 ALP3 (WIP) on the SBSA workstream wiki
  - OCP SBSA 8.1 REL targeting end of CY 2026

OCP SBMR CLA Workstream Members

ami

AMPERE.

arm

ASPEED

Google Cloud

GRAPHCORE

insyde

Meta

Microsoft

NVIDIA

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SBMR Specification Update

- **Arm specifications**
  - SBMR 1.0 (Jan 2020)
  - SBMR 2.1 (Oct 2024) → Most existing Arm-based servers
  - Current public release: Arm SBMR 3.0 ALP2 (Oct 2025) https://developer.arm.com/documentation/den0069/latest/
- **OCP Specifications**
  - OCP SBMR 3.0 ALP4 (WIP) on the SBMR Workstream Wiki
  - OCP SBMR 3.0 REL targeting end of CY 2026

Additional Resources

- **Specifications**
  - SystemReady Requirements Specification (SRS): https://support.arm.com/documentation/den0109/
  - System Standards
    - BSA, PC-BSA, SBSA (historical), SBMR (historical): https://developer.arm.com/architectures/system-standards
    - OCP SBSA: https://github.com/opencomputeproject/ocp-sbsa/wiki
    - OCP SBMR: https://github.com/opencomputeproject/ocp-sbmr/wiki
  - Base Boot Requirements
    - BBR/SBBR, BBSR, EBBR: https://developer.arm.com/architectures/base-boot-requirements
- **Architectural Compliance Suite (ACS)**
  - sysarch-acs
    - Architecture-level compliance tests https://github.com/ARM-software/sysarch-acs
  - bbr-acs and sbmr-acs
    - Firmware and security compliance tests: https://github.com/ARM-software/bbr-acs
    - Server-manageability compliance tests: https://github.com/ARM-software/sbmr-acs
  - SystemReady ACS
    - Turn-key ACS tests required for SystemReady compliance: https://github.com/ARM-software/arm-systemready

Call to Action

- Let’s continue SBSA and SBMR specifications for Arm-based servers in OCP.
- Get involved by joining the new CLA workstreams:
  - **OCP SBSA for Arm-based Servers Workstream**, under the Server Project
  - **OCP SBMR for Arm-based Servers Workstream**, under the HW Management Project
- Who should join?
  - Existing active Arm ecosystem partners in SystemArchAC
  - Other interested parties should talk to Arm about possible membership
- For more information, visit the workstreams public Wiki pages
  - https://github.com/opencomputeproject/ocp-sbsa/wiki
  - https://github.com/opencomputeproject/ocp-sbmr/wiki

Thank You!

